

Lab03 - ME315

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```
library(readr)
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)
library(stringr)
library(leaflet)
```

Questão 1

```
airports <- read_csv("airports.csv",
                     col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE))
```

Rows: 322 Columns: 5

-- Column specification -----

Delimiter: ","

chr (3): IATA_CODE, CITY, STATE

dbl (2): LATITUDE, LONGITUDE

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
airlines <- read_csv("airlines.csv")
```

Rows: 14 Columns: 2

-- Column specification -----

Delimiter: ","

chr (2): IATA_CODE, AIRLINE

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
processa_bloco <- function(bloco, pos) {
  bloco %>%
    filter(!str_starts(DESTINATION_AIRPORT, "1")) %>%
    drop_na() %>%
    group_by(DESTINATION_AIRPORT) %>%
    summarise(soma_atraso = sum(ARRIVAL_DELAY), n_voos = n(), .groups = "drop")
}

estatisticas_parciais <- read_csv_chunked(
  "flights.csv.zip",
  callback = DataFrameCallback$new(processa_bloco),
  chunk_size = 1e6,
  col_types = cols_only(
    DESTINATION_AIRPORT = col_character(),
    ARRIVAL_DELAY = col_double()
  )
)

flights <- estatisticas_parciais %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(
    atraso_medio = sum(soma_atraso) / sum(n_voos),
    n_voos = sum(n_voos),
    .groups = "drop"
  )
```

Questão 2

```
# A chave na tabela 'flights' é 'DESTINATION_AIRPORT'
# A chave na tabela 'airports' é 'IATA_CODE'

anti_join(flights, airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))
```

```
# A tibble: 0 x 3
# i 3 variables: DESTINATION_AIRPORT <chr>, atraso_medio <dbl>, n_voos <int>
```

```
flights <- flights %>%
  left_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))
```

Questão 3

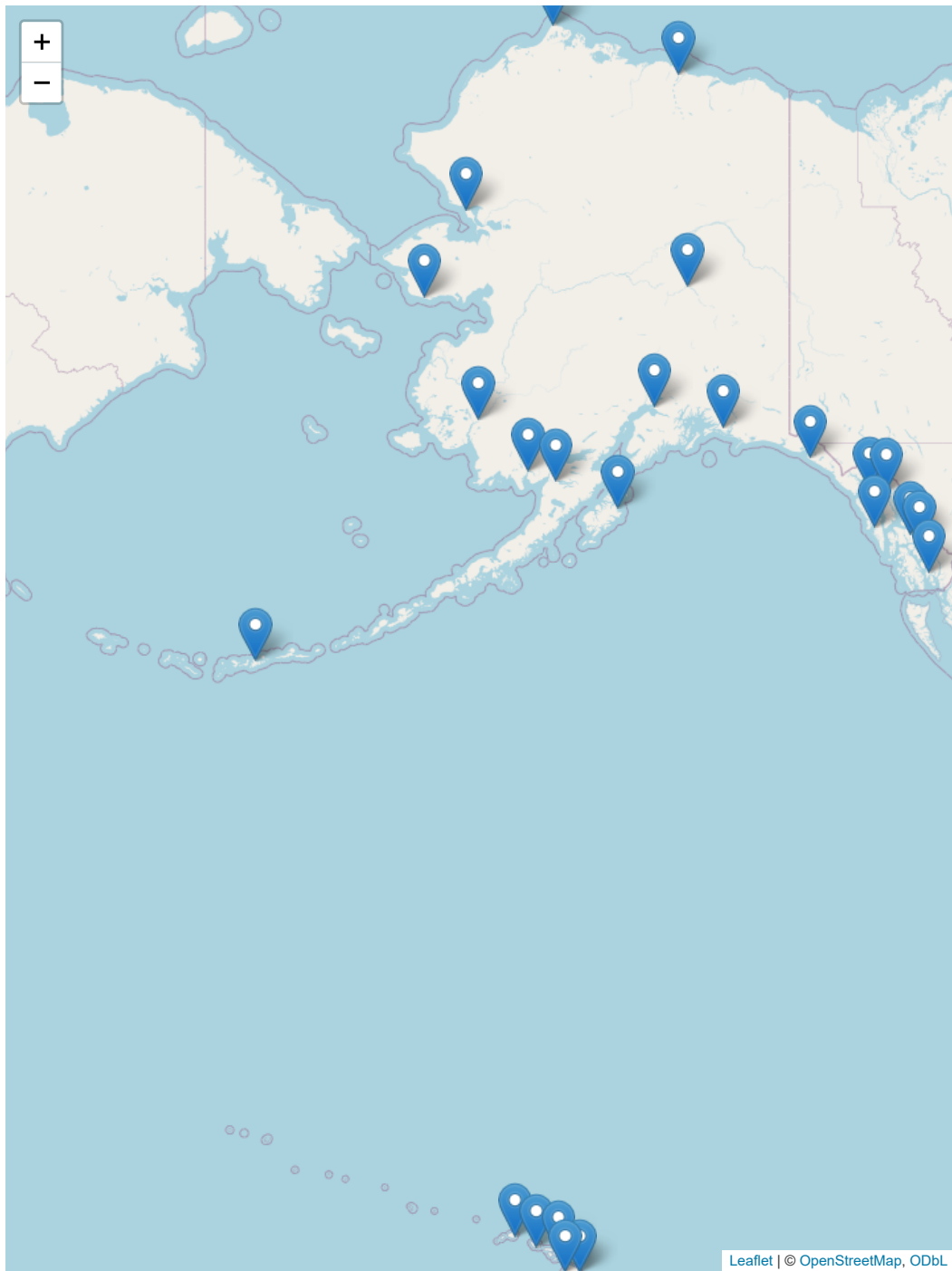
```
airports %>%
  count(STATE, name = "n_aeroportos") %>%
  arrange(desc(n_aeroportos))
```

```
# A tibble: 54 x 2
  STATE n_aeroportos
  <chr>      <int>
1 TX          24
2 CA          22
3 AK          19
4 FL          17
5 MI          15
6 NY          14
7 CO          10
8 MN           8
9 MT           8
10 NC           8
# i 44 more rows
```

Questão 4

```
dados_mapa <- flights %>%
  drop_na(LATITUDE, LONGITUDE)

this <- leaflet(data = dados_mapa) %>%
  addTiles() %>%
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~paste0(
      DESTINATION_AIRPORT, " - ", CITY, ", ", STATE, "<br>",
      "Atraso médio: ", round(atraso_medio, 1), " min<br>",
      "Voos: ", n_voos
    )
  )
this
```



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-08-29 20:44:57 -03"
```